ENGINEERING FEASIBILITY STUDY

Electric Tilting Vehicle Platform

Historical concept baseline: December 2019

Portfolio case study covering vehicle architecture, packaging, chassis engineering, FEM, ergonomics and preliminary industrialisation assessment.

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1. Concept and vehicle architecture

The project investigated a compact, fully electric, four-wheel tilting vehicle with tandem seating and a modular body architecture. Open and roofed configurations were assessed, including alternative passenger-seat and restraint arrangements. The reference electric powertrain used a 45–52 kW peak-power motor, 110–157 Nm peak torque and an approximately 14.4 kWh integrated battery concept. These are concept-stage reference values.

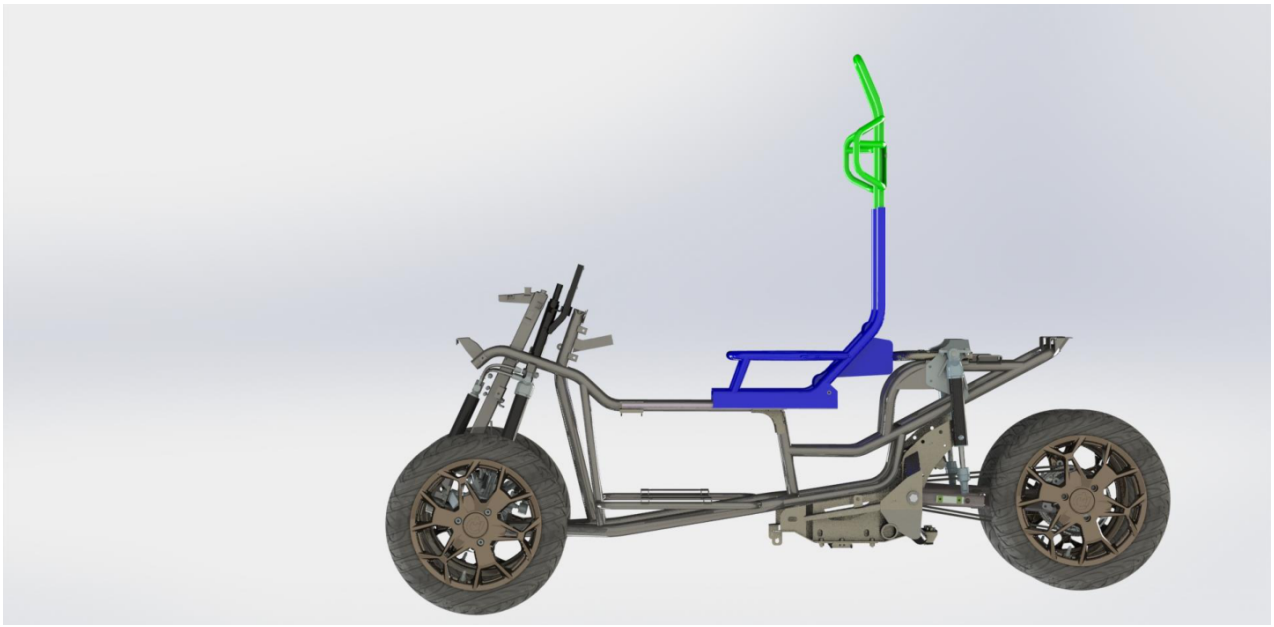
Open and roofed concept renderings

Exterior design alternatives

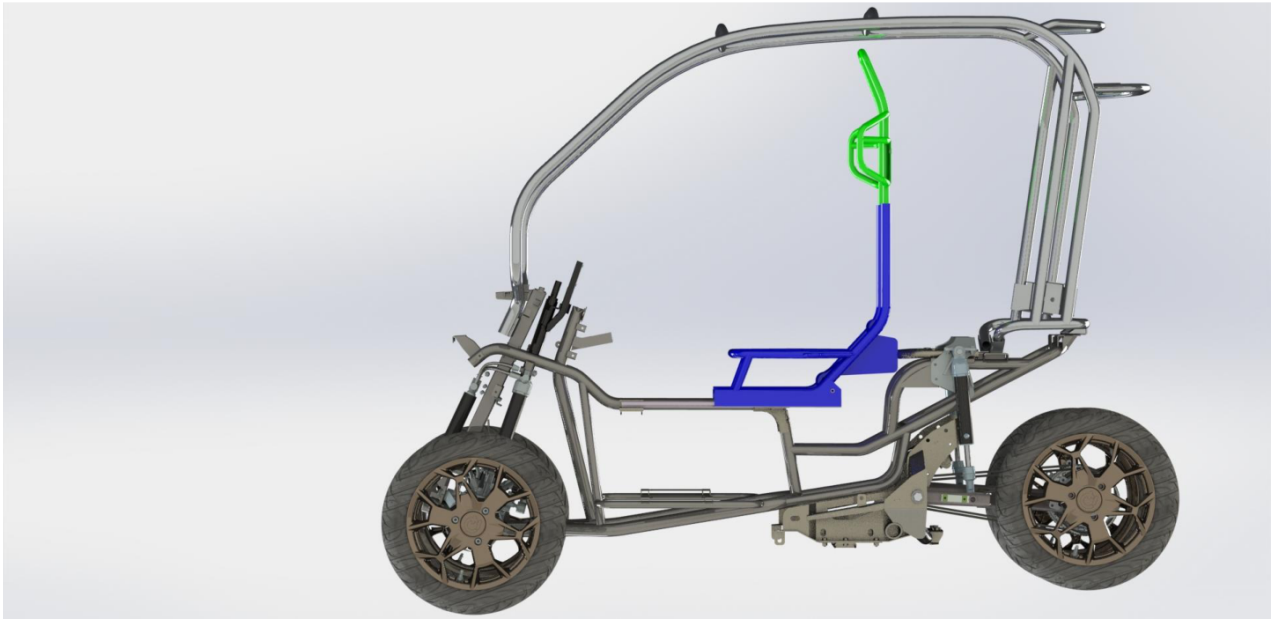
2. Modular chassis and body integration

The chassis concept was designed to support different body and seating configurations. The integration work considered roof supports, fixing points, seat positioning, sliding-seat packaging and component interfaces.

Chassis modularity – open configuration



Chassis modularity – structural seat

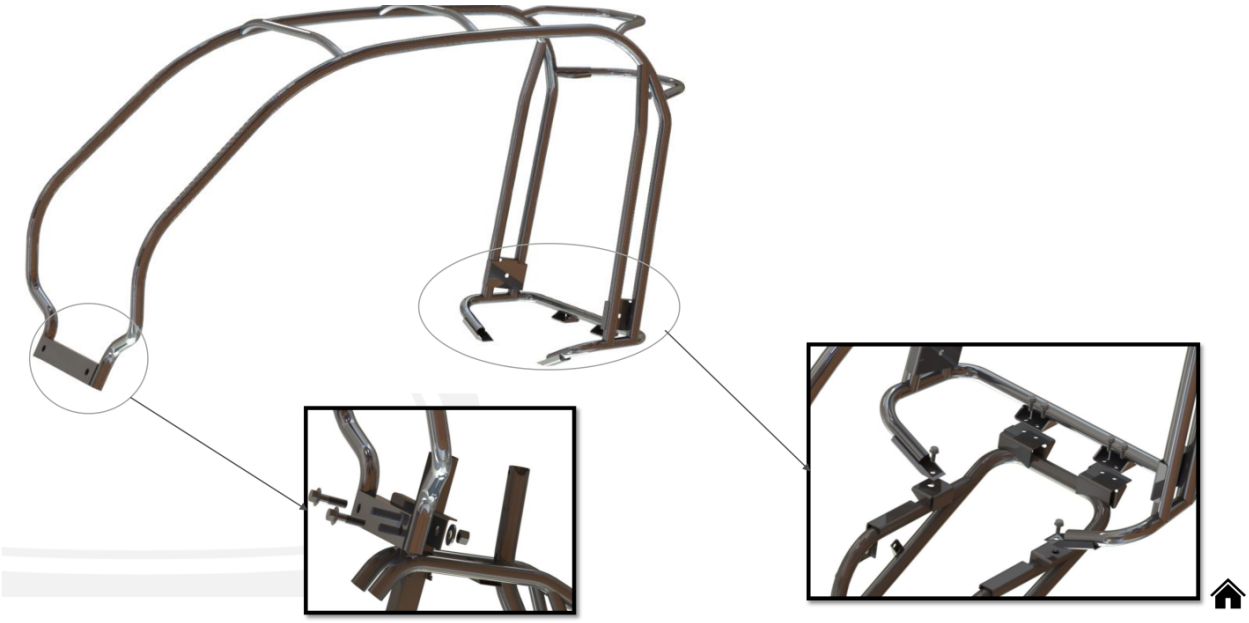


Chassis modularity – roofed configuration

Roof structure and fixing-point concept

Sliding-seat packaging

Vehicle architecture and packaging overview

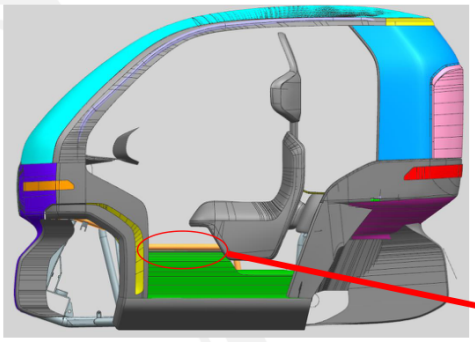


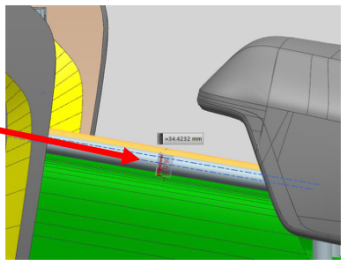
Upper-frame integration

3. System breakdown and packaging feasibility

The visual bill of materials decomposed the vehicle into chassis, suspension, steering, powertrain, braking, electrical, bodywork, seating, roof and cargo systems. The body/frame review recorded 34.4 mm nominal clearance; the rear-wheel dynamic-envelope review recorded a maximum conflict of 112.5 mm requiring resolution. Battery options were compared against frame changes, wheelbase, mass, body modifications, cable routing and available volume.

Vehicle visual bill of materials and system breakdown

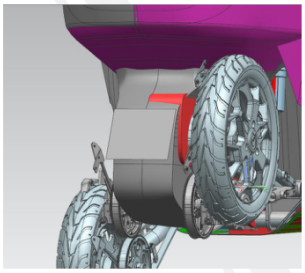


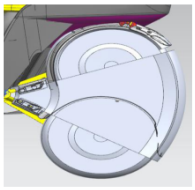


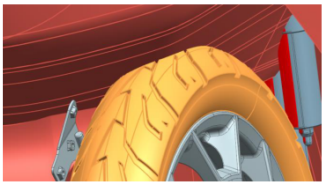
INTERFERENCE FRAME-SHELL= 34.4mm

Body/frame interference review

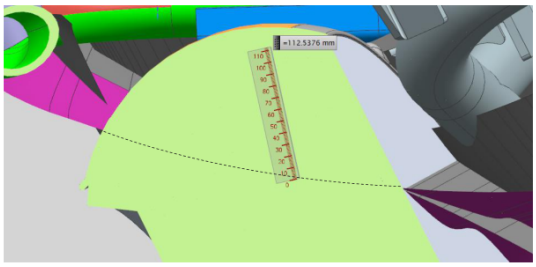
REAR WHEEELS ENVELOPE



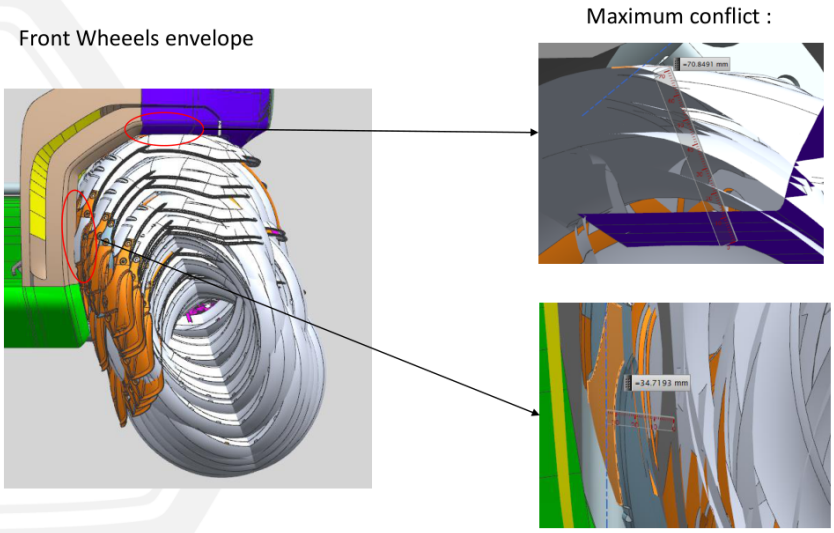




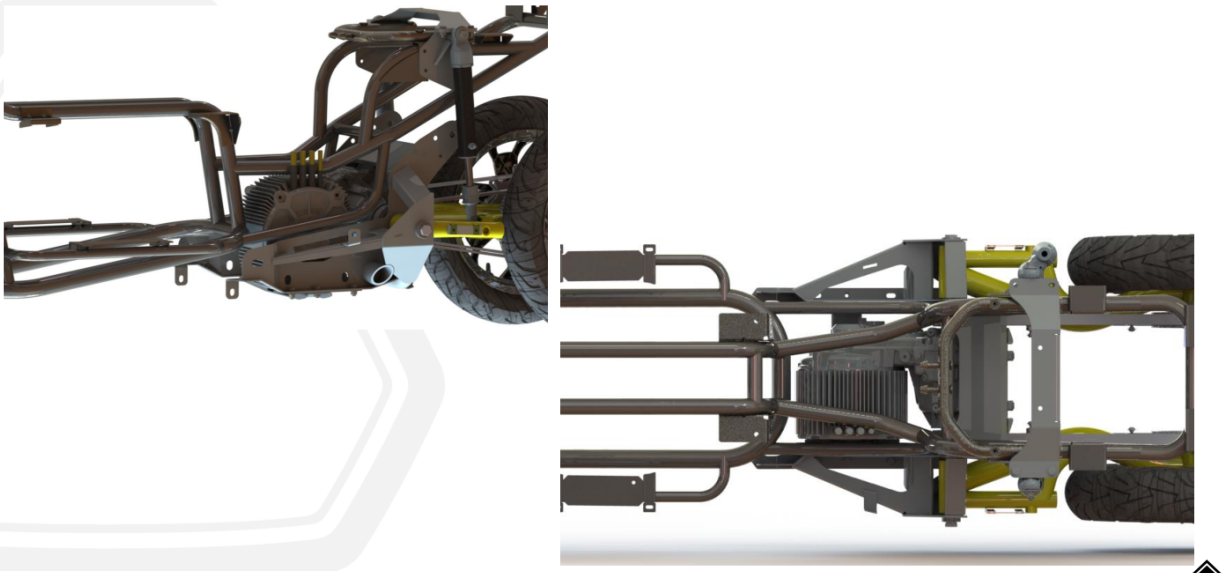
MAXIMUM CONFLICT : 112.5MM



Rear wheel dynamic-envelope check



Front wheel dynamic-envelope check



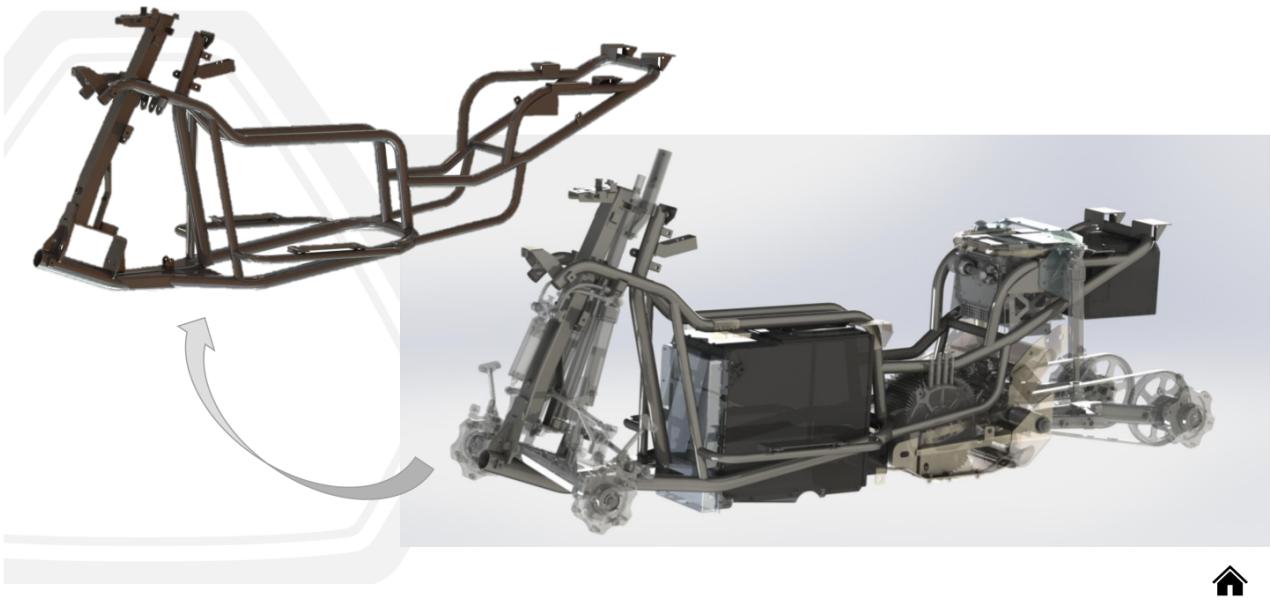
Powertrain integration and packaging

MATRIX: LAYOUT-FEASIBILITY

Assetto Motore	Batteria	Posizione	Modifica Frame	Incremento interasse Frame	Posizionamento rispetto al fondo del Telaio(mm)	Modifica Carrozzeria	Massa Batterie (P=20kg; L=40kg; M=80kg)	Vantaggi
Longitudinale	Longbrick	Orrizontale contenuta nel Frame	no	0	80	no	40kg	_Ampi vani liberi
	Longbrick + Piccola	Orrizontali sovrapposte	no	0	-50	Si, alla base di 27mm	60kg	_Vano anteriore e sedile passegero _Masse sospese concetrate al suolo
		Longbrick Orrizontale contenuta nel Frame, Piccola sedile passengero	no	0	80	no	60kg	_Ampi vani liberi
	Monolite	Orrizontale contenuta nel Frame	si	53	-35	si	80kg	_I cavi batteria potrebbero correre lungo la scocca batteria _Masse sospese concetrate al suolo
	Monolite + Lomgbrick	Monolite Orrizontale contenuta nel Frame, Piccola sedile passengero	si	0	-35	si	120kg	_I cavi batteria potrebbero correre lungo la scocca batteria _Masse sospese concetrate al suolo
Verticale	Longbrick + Piccola	Contenute nel frame	no	0	-50	Si, alla base di 27mm	60kg	_Copatto _Ampi spazi liberi _Masse sospese concetrate al suolo
	Monolite	Verticale all'interno del Frame	si	53	-28	si, sporge di 44mm	80kg	_Possible restringimento del telaio nella zona frontale _Spazio libelo per accogliere elettronica dentro il Frame _Masse sospese concetrate al suolo _Telaio accorciabile
		Orrizontale all'interno del Frame	si	53	-28	si, sporge di 41mm	80kg	_Possible restringimento del telaio nella zona frontale _I cavi batteria potrebbero correre lungo la scocca batteria _Spazio libelo per accogliere elettronica dentro il Frame _Masse sospese concetrate al suolo _Telaio accorciabile
	Monolite + Piccola	Monolite dentro il Frame Piccola sedile passeggero	si	53	-28	si, sporge di 44mm	100kg	_Possible restringimento del telaio nella zona frontale _Autonomia veicolo _Spazio libelo per accogliere elettronica dentro il Frame _Telaio accorciabile
		Contenute nel frame	si	53	-28	si, sporge di 44mm	100kg	_Possible restringimento del telaio nella zona frontale _Masse sospese concetrate al suolo
	Monolite + Longbrick	Contenute nel frame	si	53	-28	si, sporge di 65mm	120kg	_Comaptto _Masse sospese concetrate al suolo

Battery layout feasibility matrix

Battery pack layout alternatives



Central frame design

4. Structural engineering and FEM

The lower frame was assessed as a welded steel structure (2 mm nominal thickness; 36SMn14 in the source). The roof structure was assessed in aluminium alloy 7050-T73510 (3 mm nominal thickness). Load cases included vertical frame loading, 1 g braking, 100 Nm torsion, upper-body vertical/transversal loading and driver-seat vertical/transversal loading. Reported frame torsional stiffness was approximately 11,792 Nm/°. Seat-slide areas showed local stress concentrations; increasing fasteners was recommended. Rollover assessment was still in progress in December 2019.

Lower frame preliminary FEM model

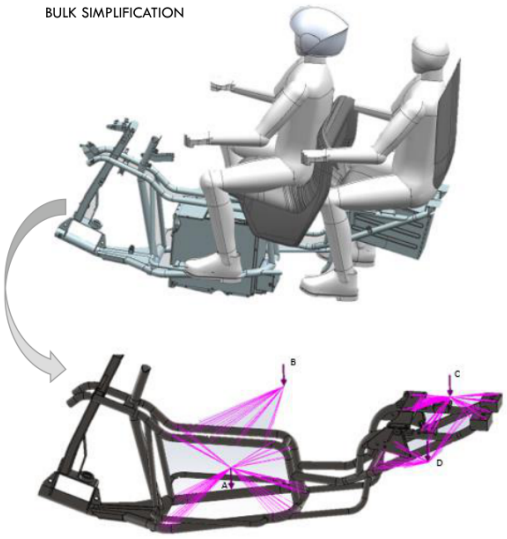
A: BATTERY MONOLITH: 80KG

B: 1ST PASSENGER: 80KG

C: 2ND PASSENGER: 80KG

D: BATTERY MODULAR + 6kW CHARGER: 40KG

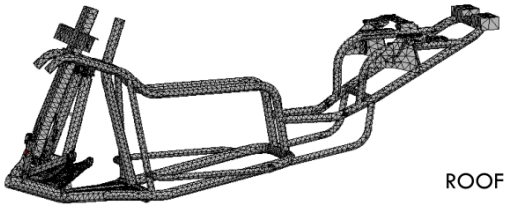
BULK SIMPLIFICATION



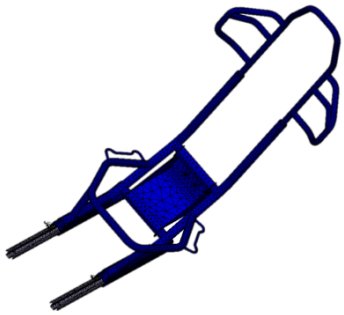
FEM mass and load simplification

MESH

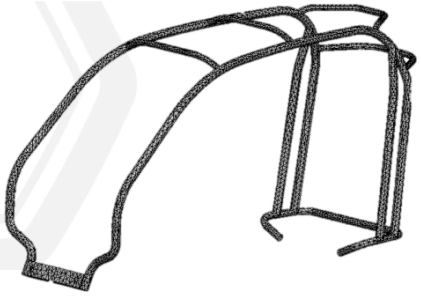
MAIN FRAME



SEAT



ROOF

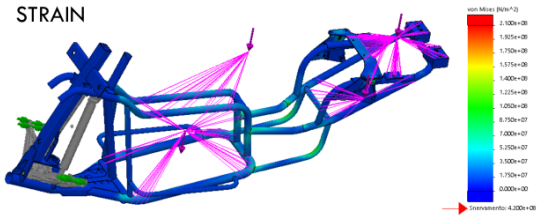


Lower frame FEM mesh

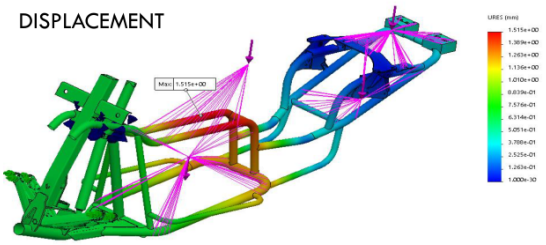
FRAME: VERTICAL LOAD

AS CONVENTION, THE MAXIMUM STRESS IS SETTLED FOF HALF YELD STRENTH: YS/2

STRAIN



DISPLACEMENT



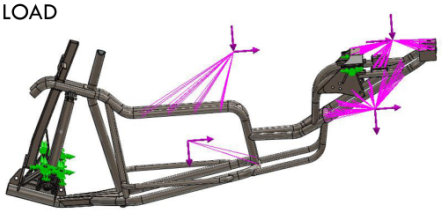
GOOD MODEL RESPONSE

FEM – frame vertical load

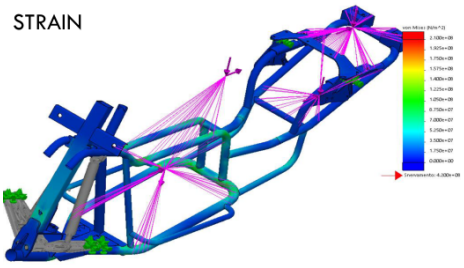
FRAME: BRAKE LOAD

BY ACCELERATION OF 1G

LOAD



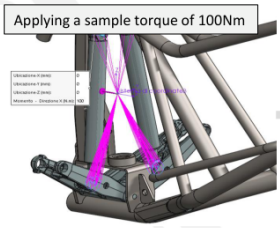
STRAIN

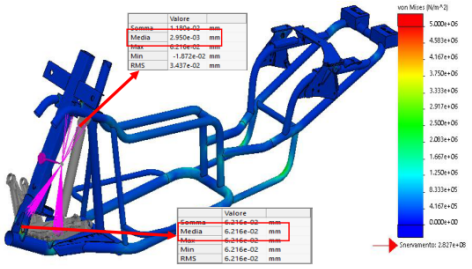


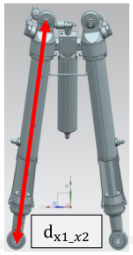
GOOD MODEL RESPONSE

FEM – frame braking load

FRAME: TORQUE







TORQ_DEF= 100NM

X1_DEF= 6.216E-02

X2_DEF= -2.95E-03

D_{X1_x2} = 440MM

$\theta_DEF = \frac{X1_{DEF} - X2_{DEF}}{D_{X1_x2}} * \frac{180}{\Pi} = 0.00848°$

$KTORS_KOMMA = \frac{Torq_{def}}{\theta_{def}} = 11'792.45\ Nm/°$

KTORS_NUVION= 693.1 NM/° → KOMMA FRAME IS MORE STIFF!

GOOD MODEL RESPONSE

FEM – frame torsional stiffness

UPPER BODY: VERTICAL LOAD

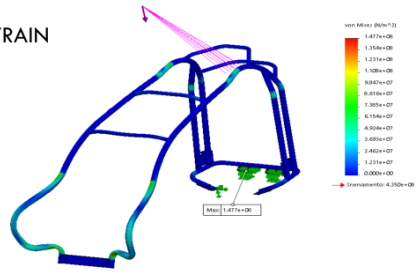
$M_{TEST} = M_{EQOODER} + \Delta M_{BATTERIES} + 2*M_{PASSENGER} + \Delta M_{FRAME}$

$M_{EQOODER}$= 293kg
$\Delta M_{BATTERIES}$= 60kg
$M_{PASSENGER}$= 80kg
ΔM_{FRAME}= 20kg

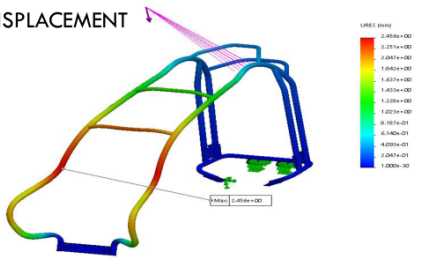
M_{TEST}= 533kg
$F_V = 2G*M_{TEST} \approx$ 10.400N

GOOD MODEL RESPONSE

STRAIN



DISPLACEMENT

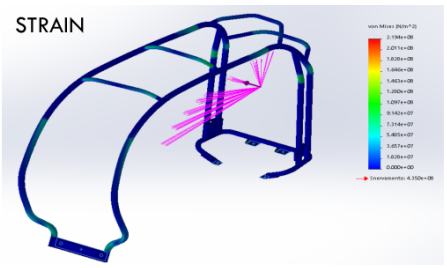


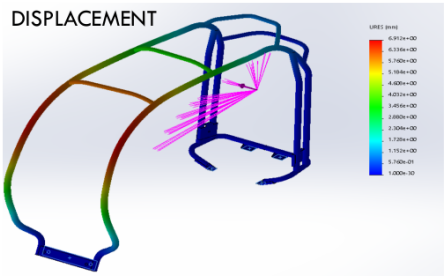
FEM – upper body vertical load

UPPER BODY: TRANSVERSAL LOAD

M_{TEST}= 533kg
F_V= G* M_{TEST} ≈ 5.600N

GOOD MODEL RESPONSE



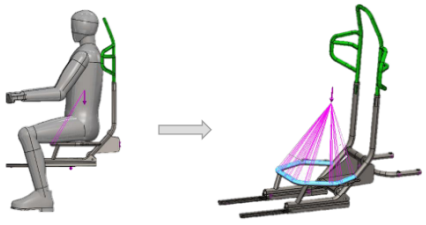


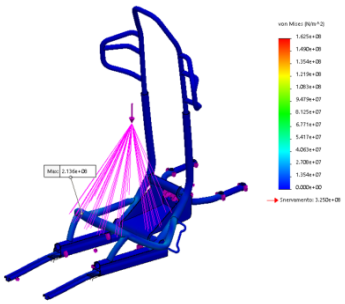
FEM – upper body transversal load

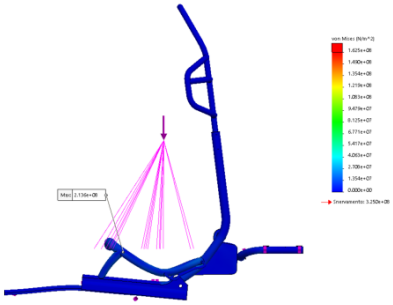
DRIVER SEAT: VERTICAL LOAD

M_{DRIVER}= 80kg
ACCELERATION= 2.5G

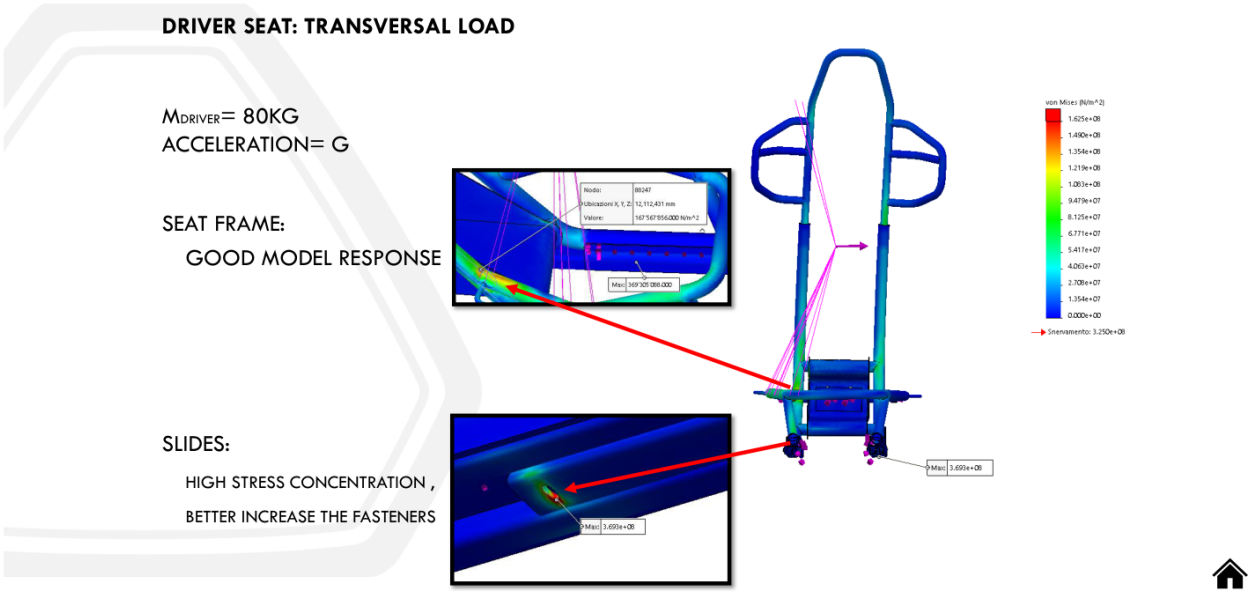
GOOD MODEL RESPONSE



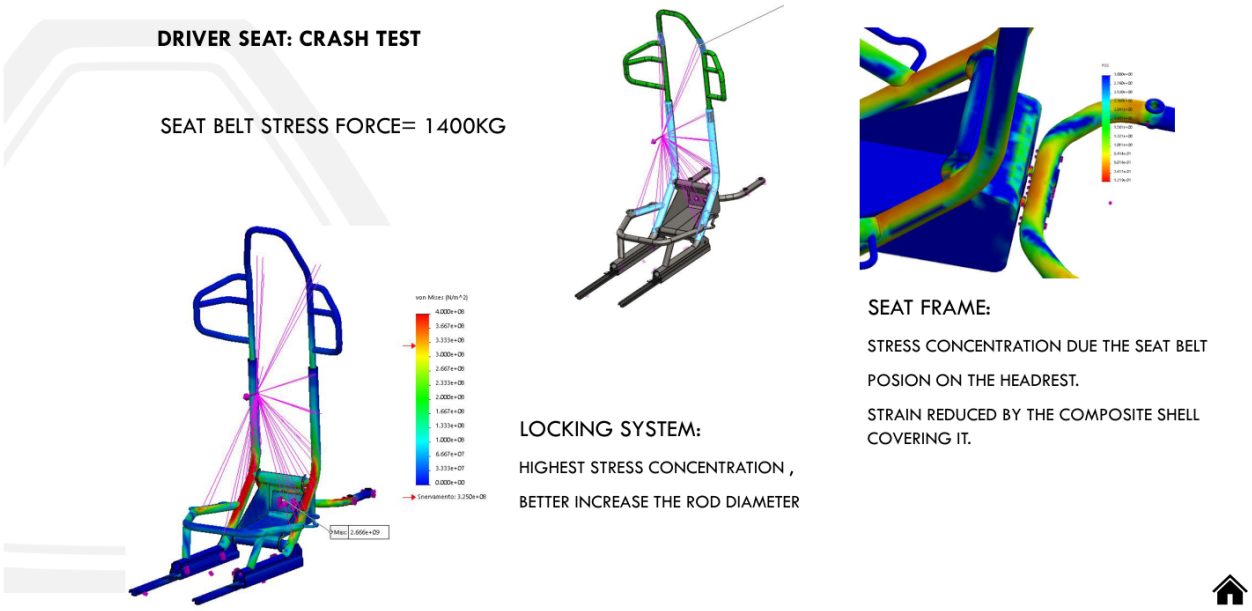




FEM – driver seat vertical load



FEM – driver seat transversal load



Rollover protective structure assessment status

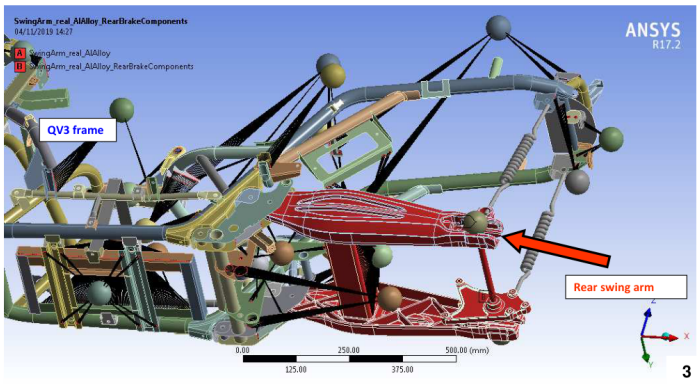
5. Rear swing-arm component FEM

A separate aluminium-alloy rear swing-arm study documented the FEM model, load cases, two load-condition result sets and torsional-stiffness assessment. These component-level results should not be conflated with the vehicle-level frame FEM results.

1. FEM MODEL

1.1) Introduction

The aim of this report is to verify the Aluminium Rear swing arm version on EQV3 vehicle. For other details about FEM model (i.e. constraints) see rep.771.

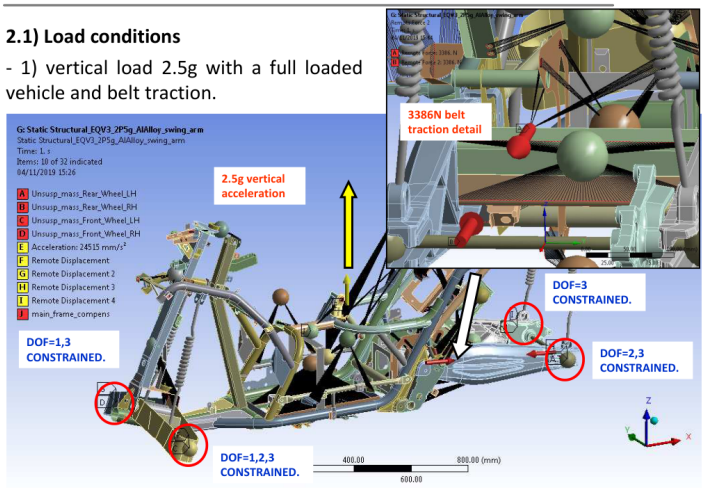


Swing-arm FEM model

2. LOADS

2.1) Load conditions

- 1) vertical load 2.5g with a full loaded vehicle and belt traction.

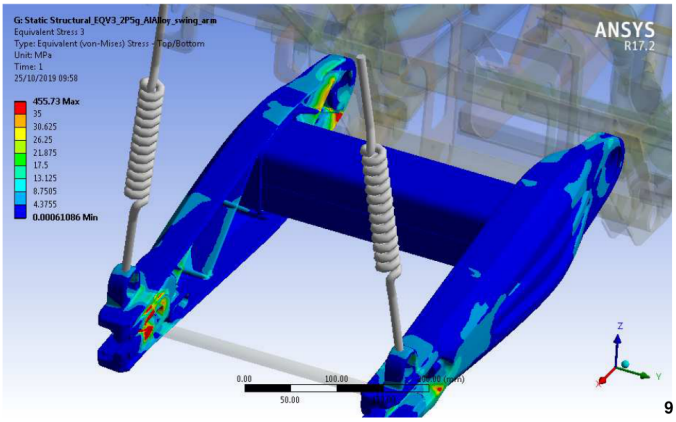


Swing-arm FEM load cases

3. RESULTS LOAD CONDITION 1

3.2) Equivalent Von Mises stress

Al Alloy swing arm:

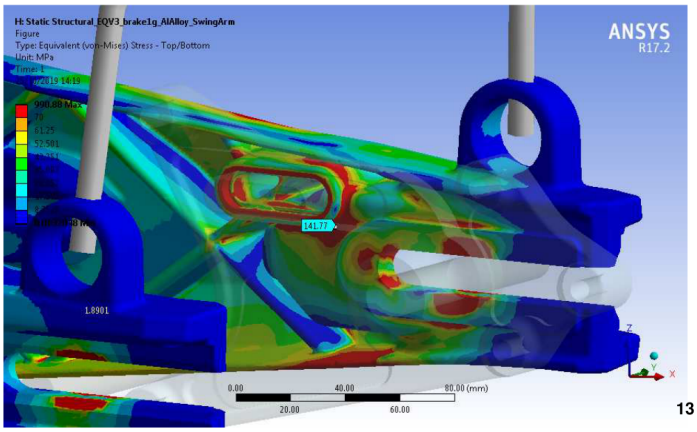


Swing-arm FEM results – load case 1

4. RESULTS LOAD CONDITION 2

4.2) Equivalent Von Mises stress

Rear swing arm:

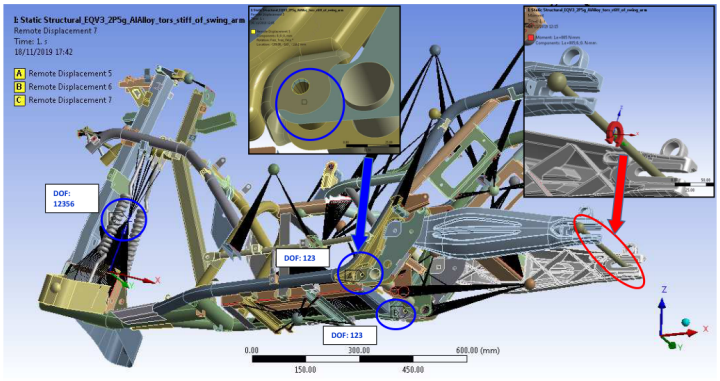


Swing-arm FEM results – load case 2

5. TORSIONAL STIFFNESS

5.1) Load conditions and constraints

The constraints and the sample torque of 100Nm are showed below, the DOF constrained are indicated:



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Swing-arm torsional stiffness

6. CONCLUSION

1. **The link from caliper support and swing arm have to be modified.**

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Swing-arm FEM conclusion

6. Ergonomics and occupant packaging

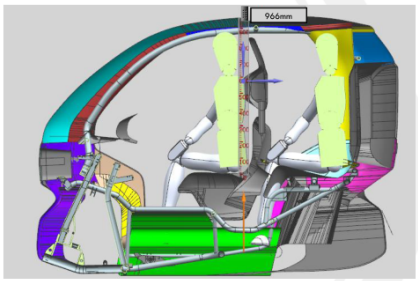
The roofed-body study evaluated helmet clearance and driver accommodation. Available internal height was estimated at approximately 913 mm. A 1,784 mm driver-height reference and 50 mm helmet-to-roof clearance corresponded to approximately the 88.3rd percentile in the source population model.

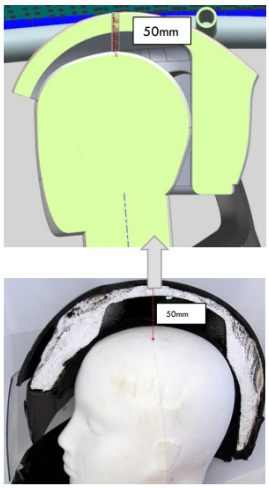
AVERAGE HELMETS THICKNESS= 50MM

GLASS THIKNESS= 5MM → ASSUMING SHELLS AS CENTER LINES= 2.5MM

REDUCING HEIGHT GAP ≃ 53MM

HEIGHT AVAILABLE= 966 – 53 ≃ **913MM**





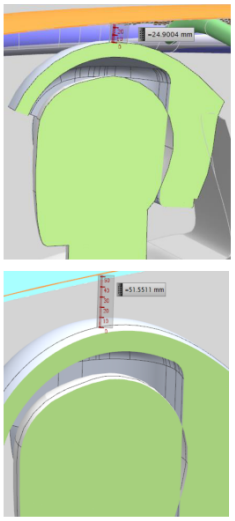
Ergonomic packaging and helmet clearance

DRIVER

TAKING ON THE 95% OF HEIGHT POPULATION → 1784MM
25MM OF CLEARANCE→ NOT ENOUGH

ASSUMING 50MM HELMET-ROOF CLEARANCE → MAXIMUM FITTING GAP 1750MM

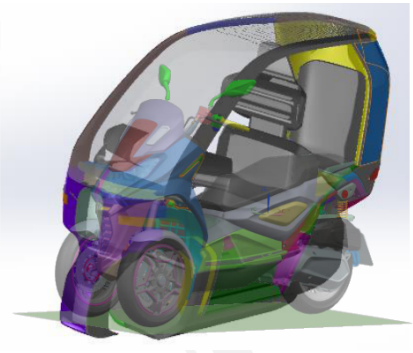
INTERPOLING THE GAUSS DISTRIBUTION → **PERCENTILE= 88,3**



Driver accommodation percentile assessment

7. Mass and component complexity

The source comparison recorded 21 versus 13 body-plastic components and 28 versus 12 structural-frame components. Reported masses were 25.6 versus 5.75 kg for body plastics and 54 versus 22.5 kg for the frame. Source percentage annotations are inconsistent; absolute values should be verified before drawing firm conclusions.



COMPARISON TABLE

	WEIGHT (kg)	D WEIGHT %	STRUCTURAL COMPONENTS
NUVION	5.7		13
KOMMA	25.6	+ 345%	21

Bodywork component complexity

COMPARISON TABLE

	WEIGHT (kg)	D WEIGHT %	STRUCTURAL COMPONENTS
NUVION	22.5		12
KOMMA	54	+ 140%	28

Frame component complexity

8. Programme status, risks and next steps

At the December 2019 baseline, major packaging, frame design and several FEA activities were recorded as complete. Seat/belt integration, windscreen/wiper integration, aerodynamic CFD and rollover protection remained in progress. Key closure activities included resolving wheel dynamic envelopes, freezing battery layout and HV routing, completing rollover/fatigue/joint validation, closing seat-slide and restraint issues, validating glazing and aerodynamic loads, completing homologation assessment, obtaining supplier quotations and rebuilding the business case with supplier-backed data.

9. Preliminary business case

The historical base case assumed 15,000 units over five years, average retail price of €28,400 and net revenue of €20,202 per unit. It reported €5.87 million R&D;, €5.02 million tooling in the summary case, €10.89 million total investment, 2,791-unit break-even, 1.2-year payback and €41.3 million NPV. The detailed tooling table reports €4.02 million, conflicting with the €5.02 million summary value. These are historical model outputs based on estimates, not

independently audited results; supplier quotations were not available and OEM production tooling was excluded.

10. Overall feasibility conclusion

The concept demonstrated a credible engineering pathway to a prototype and detailed-validation phase. The available material does not establish production readiness, homologation or commercial validation. The remaining critical path included dynamic clearances, rollover protection, occupant restraints, aerodynamic validation, supplier confirmation, physical testing and homologation.

Source and confidentiality note

This portfolio edition removes company/product naming from the report text and branded slide headers. The source figures and images are derived primarily from the December 2019 feasibility presentation, supplemented by a rear swing-arm aluminium-alloy FEM report and related technical material. Confirm ownership and publication rights for all images, CAD screenshots, supplier references and engineering data before public release.